

The operating window for copper–molybdenum separation: a Gaussian process model of concentrate grade, and of what it does not control

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Abstract

A variogram-based Gaussian process model is fitted to twenty-nine bench flotation tests from a copper–molybdenum concentrator and inverted for the operating conditions that minimise the copper penalty in the molybdenum concentrate. Two points of method are settled first. The campaign is not one-factor-at-a-time but three two-factor planes sharing a grind-fineness axis, so a kriging surface is recoverable on each; and the response is concentrate grade rather than recovery, because the archived grades satisfy the mass balance throughout while the archived copper recoveries do not. Both grades are predictable from grind fineness, pulp pH, diesel dosage and conditioning time, and the surfaces recover an interaction between fineness and diesel dosage that a single-length-scale model cannot represent. Conditioning time is inert. The inversion gives an operating window at the upper bounds of the tested ranges, corroborated by the nearest measured test and by resampling of the data. A limitation of the study is that none of the four operating variables correlates strongly with the separation, whereas pulp rheology and concentrate wettability vary strongly with clay content and temperature, the correlation between wettability and temperature changing from negative to positive when clay is present.

Keywords: froth flotation, copper–molybdenum separation, Gaussian process regression, kriging, concentrate grade

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1. Introduction

Copper in a molybdenum concentrate is a smelter penalty that cannot be removed downstream. Molybdenite and chalcopyrite are both naturally hydrophobic, so reagents that float one also float the other, and the separation rests on small differences in floatability. A plant can adjust few variables: grind fineness, pulp pH, collector dosage and conditioning time. A campaign that varies these and assays the concentrate supports a response surface from which untested settings can be proposed.

Gaussian process regression is suited to this, being defined on small datasets and returning a posterior variance as part of its output. Eskanlou, Yin and Caers derive the prior covariance function from empirical variograms in the space of the operating variables and demonstrate the approach on a designed flotation campaign, mapping a Pareto-efficient frontier back to process settings. This study applies that framework to a plant campaign of a different character, and reports both what the resulting surface supports and what it does not.

2. Data

Twenty-nine rougher flotation tests were run at a molybdenum concentrator at Jinduicheng, Shaanxi Province, on a 300 g feed assaying 0.5436 % molybdenum and 0.0364 % copper. Grind fineness ranged from 30 to 65 % passing 0.074 mm, pulp pH from 3.5 to 11, diesel dosage from 50 to 200 g/t, and diesel–pulp conditioning time from 5 to 25 min. Molybdenum grade in the rougher concentrate ranged from 1.10 to 6.67 % and copper grade from 0.215 to 0.834 %.

Each reagent was swept from the same two anchor values of grind fineness, 40 % and 65 %, and the fineness sweep was run at the anchor values of the reagents. The design is therefore three two-factor planes sharing a fineness axis — fineness against pH, against diesel dosage and against conditioning time, with fifteen, fifteen and seventeen tests respectively — and fineness–reagent interactions are identified (Fig. 1). Reagent–reagent interactions are not sampled.

The response is the concentrate grade rather than the recovery. With a 300 g feed at 0.5436 % molybdenum, concentrate mass multiplied by molybdenum grade reproduces the reported molybdenum recovery in all twenty-nine tests; the copper recovery is reproduced in fifteen, and therefore depends on a feed assay not archived with the test. Grade is a direct measurement of the concentrate. A second check against the figures published

from the same archive leaves copper grade unchanged in all twenty-nine tests and differs in seven molybdenum values, in each of which the archived value satisfies the mass balance.

Three production lines were surveyed over 50 to 75 % fineness, giving eighteen grade measurements that are not used to fit the model; because all three were surveyed at the same six levels, they also give the plant pure error. Pulp viscosity was measured against temperature, shear rate, and clay identity and loading, for chlorite, illite, muscovite and kaolinite, each condition twice; contact angle was measured in a clean system and in one containing 25 % chlorite, against temperature, chlorite content and diesel dosage.

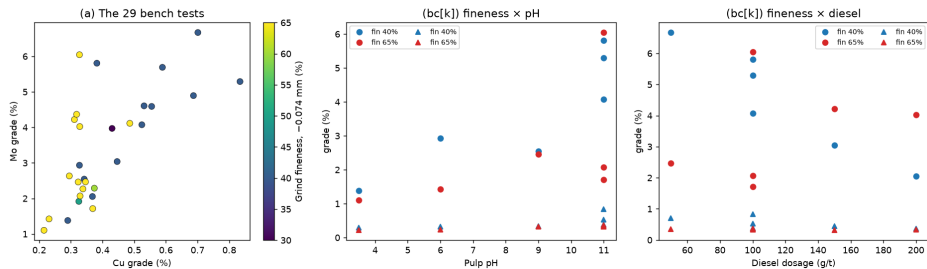


Figure 1: (a) The twenty-nine bench tests, coloured by grind fineness. (b), (c) The fineness $\times$ pH and fineness $\times$ diesel planes; markers distinguish the two grades and colours the two fineness anchors.

3. Model

Each response is modelled as a Gaussian process with an automatic relevance determination covariance function,

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_{i=1}^d \frac{(x_i - x'_i)^2}{\ell_i^2}\right) + \sigma_n^2 \delta_{xx'}, \quad (1)$$

with predictors scaled to the unit interval and all hyperparameters, including the nugget σ_n^2 , obtained by maximising the marginal likelihood. A non-zero nugget is required: flotation is not a deterministic simulator.

Fits are judged by leave-one-out cross-validation, reported as a coefficient of determination Q^2 and a root-mean-square error, and by the loss in Q^2 when one predictor is withheld. Two-dimensional surfaces are slices through the fitted four-dimensional model at the anchor values of the two factors not shown; a separate two-parameter fit on each plane cross-validates worse, with

Q^2 between -0.01 and 0.40 . An operating window is sought over 160,000 grid points. Molybdenum grade is to be raised and copper grade lowered, so the non-dominated set is the set of settings for which no other setting improves one grade without worsening the other, and a recommendation is taken from it as the point closest to the ideal. Its stability is examined by resampling the data rather than by drawing Gaussian realisations, which are not constrained to be positive.

4. Results

4.1. Prediction of grade

Both grades are predictable, copper more so than molybdenum (Table 1, Fig. 3). The fitted nugget accounts for 0.34 of the total variance of molybdenum grade and 0.43 of copper grade; the empirical variograms of Fig. 2, which do not absorb short-scale structure that the kernel cannot fit, put the unstructured fraction at 0.21 for both. Residuals are 1.16 and 0.10 percentage points against observed standard deviations of 1.56 and 0.147.

Table 1: Leave-one-out performance, and the loss in Q^2 when one predictor is withheld.

	Molybdenum grade			Copper grade		
	Q^2	RMSE	ΔQ^2	Q^2	RMSE	ΔQ^2
All four predictors	0.434	1.155	—	0.527	0.099	—
without fineness			0.287			0.453
without pH			0.380			0.265
without diesel			0.193			0.094
without conditioning time			0.000			0.000

The two criteria rank the predictors identically for molybdenum grade, where pH and fineness are comparable and diesel is third. For copper grade they disagree: the inverse length scales of Fig. 3c rank fineness last, at 0.127 against 0.675 for diesel, whereas withholding fineness costs more Q^2 than withholding anything except pH. The fineness sweep covers 30 to 65 % in five steps and contains a sharp maximum, which one length scale cannot express without shortening the others, so the length scale is not a reliable sensitivity measure for this variable. Conditioning time contributes nothing under either criterion — its length scale sits at the bound of the optimisation range and withholding it changes Q^2 by 0.000 for both responses — although grades along the time series vary by more than a factor of two. That variation is scatter.

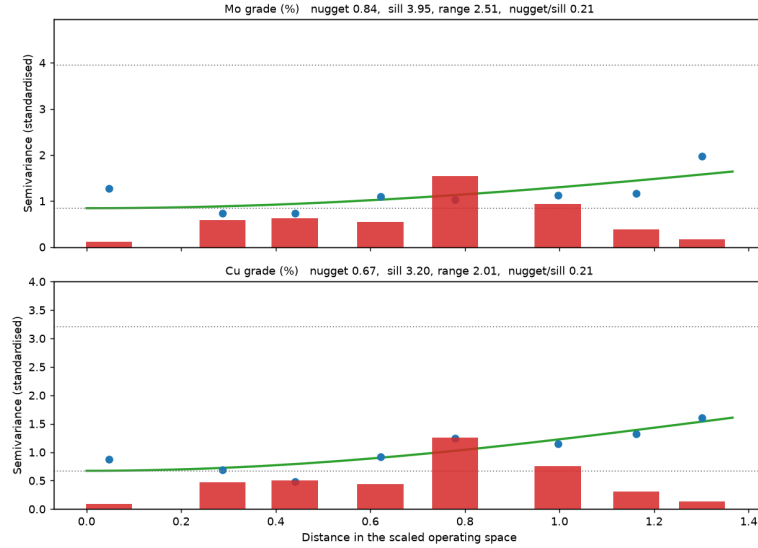


Figure 2: Empirical variograms in the scaled operating space with the fitted Gaussian models; bars give the number of pairs per distance bin.

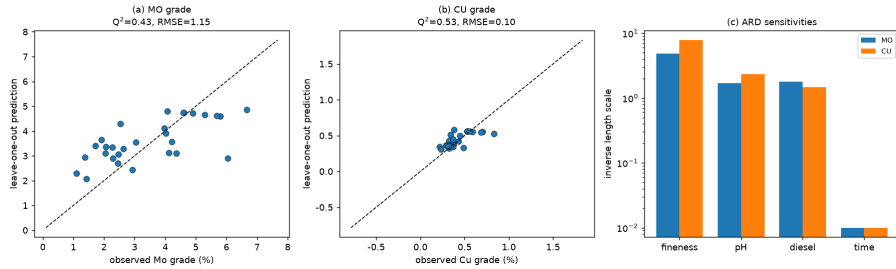


Figure 3: Leave-one-out predictions and the inverse length scales of the fitted covariance function.

4.2. The three planes

The fineness $\times$ pH plane has a single optimum at high pH and low fineness, at which both grades peak. The fineness $\times$ diesel plane contains an interaction: at the low fineness anchor, molybdenum grade falls monotonically with diesel addition, from 6.67 % at 50 g/t to 2.05 % at 200 g/t, whereas at the high fineness anchor it rises from 2.47 to 4.02 % over the same range (Fig. 4). Neither structure is representable by a single length scale per variable.

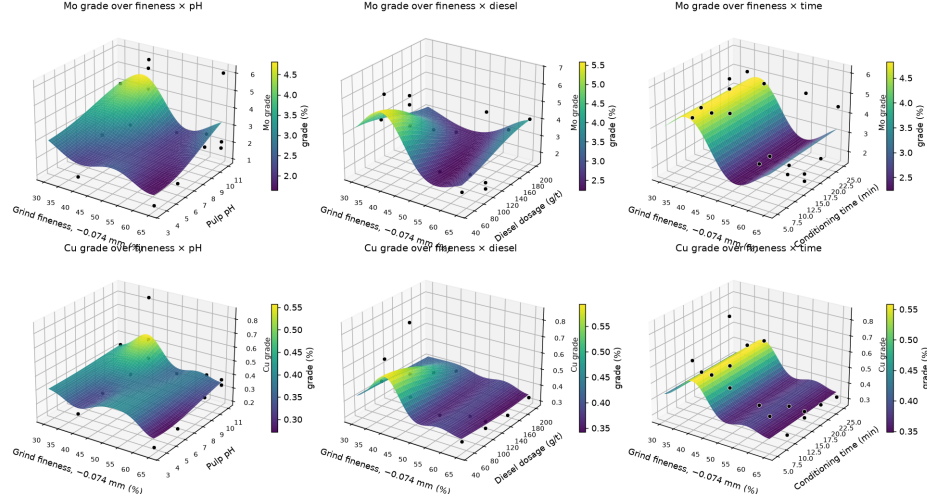


Figure 4: Fitted kriging surfaces, taken as slices through the four-dimensional model at the anchor settings of the two factors not shown. Points are the tests lying on each plane.

4.3. The operating window

The two grades correlate at $+0.678$, so the trade-off of Fig. 6a is the boundary of a cloud; the non-dominated set contains 747 of the 160,000 grid points. The recommended conditions are given in Table 2, with the mean, standard deviation and probability of meeting a copper target in Fig. 5. Both the fineness and the diesel dosage lie at the upper bounds of the tested ranges, which is why the plant surveys are needed.

Table 2: Recommended operating conditions. The first row is the point of the non-dominated set closest to the ideal; the second minimises copper grade subject to a molybdenum grade of at least 3.5 %. The third row gives the median and standard deviation of the second row over one hundred resamples of the data, and the fourth the nearest measured test. Tested ranges are 30–65 % fineness, pH 3.5–11, diesel 50–200 g/t, conditioning 5–25 min.

	Fineness (%)	Pulp pH	Diesel (g/t)	Time (min)	Mo (%)	Cu (%)
Balanced trade-off	65	11.0	200	24	3.97	0.347
Min. Cu, Mo ≥ 3.5 %	65	9.4	168	25	3.51	0.328
median of resamples	63 ± 6	9.3 ± 2.2	154 ± 28	21 ± 5	—	0.329 ± 0.041
Measured test 20	65	11.0	200	15	4.02	0.328

Three checks apply. The nearest measured test, test 20, was run at the same fineness, pH and diesel dosage with a conditioning time of 15 min and returned 4.02 % molybdenum and 0.328 % copper, so the predicted grades

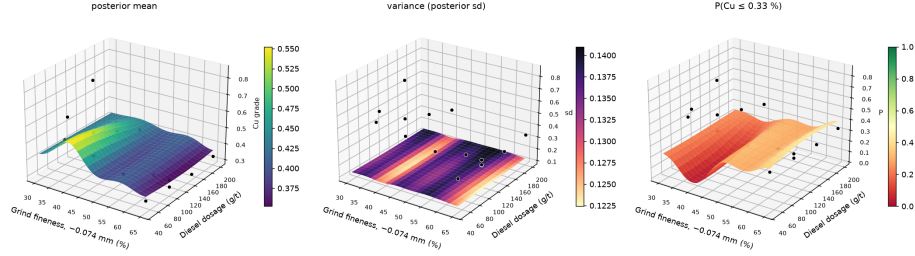


Figure 5: Posterior mean and standard deviation of copper grade over the fineness $\times$ diesel plane, and the probability that it is at or below 0.33 %.

agree with a measurement at adjacent settings to within 0.02 percentage points of copper. Refitting from eight random initialisations returns the same point. Resampling places the copper grade achievable subject to a molybdenum grade of at least 3.5 % at 0.329 ± 0.041 %. At a molybdenum target of 4 % the best achievable copper grade rises to 0.416 %, so the constraint binds.

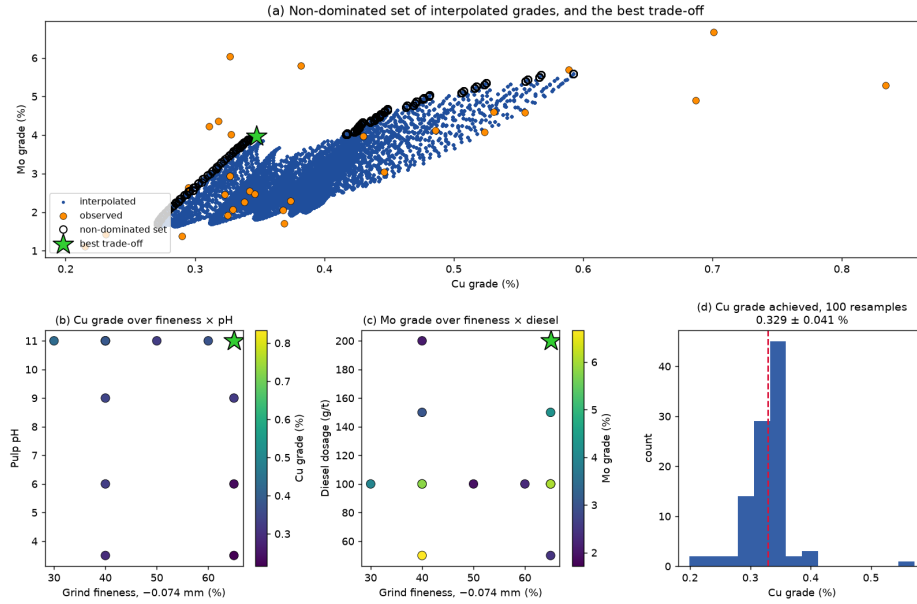


Figure 6: (a) Interpolated grades with the non-dominated set and the best trade-off. (b), (c) Projections into input space with the recommended point marked. (d) Copper grade achieved at the constrained optimum over one hundred data resamples.

4.4. Transfer to the plant

The bench design ends at 65 % fineness, so the surveys locate the optimum relative to that boundary. All three lines reach their maximum molybdenum grade at 65 % and their minimum copper grade between 70 and 75 %, and molybdenum grade falls sharply at 75 % (Fig. 7); with the bench model this places the optimum at 65 to 70 %. The three lines give a pooled standard deviation of 0.82 percentage points of molybdenum grade and 0.053 of copper grade on fifteen degrees of freedom, against a spread of 4.08 and 0.215 across the fineness range — a signal-to-noise ratio of about five for molybdenum and four for copper.

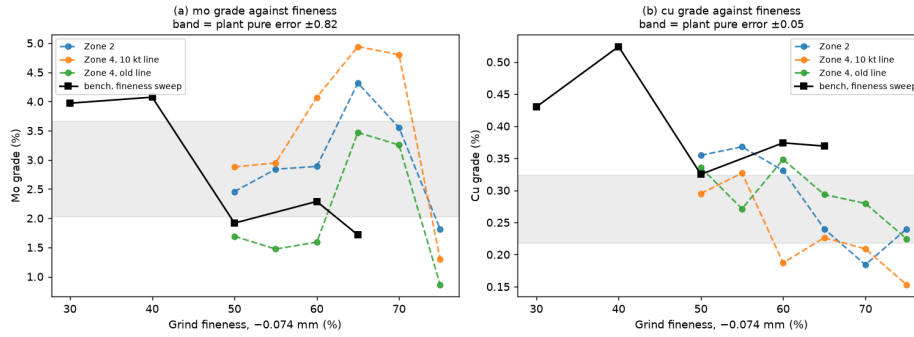


Figure 7: Both grades against grind fineness for the three plant surveys and the bench fineness sweep. The band is the plant pure error, $\pm$ one pooled standard deviation about the mean.

4.5. The pulp environment

Apparent viscosity falls with temperature for chlorite by 15.8 % over 5 to 35 °C, for muscovite by 12.3 % and for illite by 7.0 %. Kaolinite gives about 385 mPa s where the others give 180–205, and falls by only 2.5 %; its replicates differ by 32 mPa s against 5 to 9 for the others, so the correct statement is the absence of a detectable temperature response rather than a smaller one. A clean concentrate loses hydrophobicity on heating, its contact angle falling from 82.8° at 10 °C to 67.8° at 50 °C; with 25 % chlorite it gains hydrophobicity over the same interval, from 85.6° to 91.3° (Fig. 8). The sign of the temperature coefficient therefore reverses with clay present, and the reversal is established by the time a quarter of the solids are clay. Diesel dosage acts in the same direction and to a similar optimum near 100 g/t in both systems, but the effect is twice as large without clay.

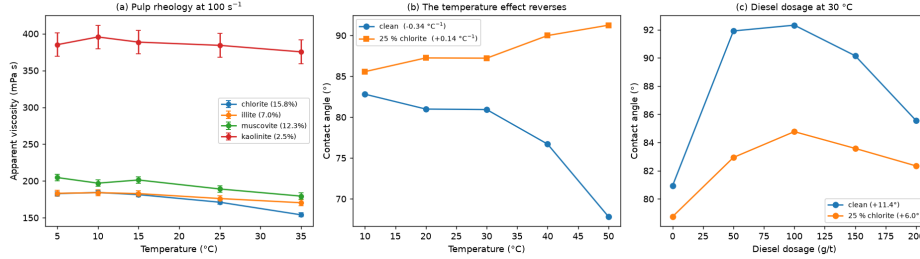


Figure 8: (a) Apparent viscosity against temperature for four clays at 100 s^{-1} , with the replicate spread. (b) Contact angle against temperature in the two systems. (c) Contact angle against diesel dosage.

5. Discussion and conclusions

The recommended window is supported by a measured test at adjacent settings, by refitting, and by the fineness optimum of three plant surveys, and should be read against the uncertainty of Fig. 5. The quantity the decision depends on, the copper grade achievable subject to a molybdenum constraint, is better determined than the settings that deliver it.

The operating variables do not control the separation. Molybdenum and copper grade correlate at $+0.678$, and their ratio is not predictable from any of the four: fitted directly to the ratio the same covariance function returns $Q^2 = -0.072$, and forming the ratio from the two leave-one-out predictions returns $Q^2 = -0.002$. Two independent routes agreeing on a negative result exclude an artefact of a particular fit, though not an effect smaller than the dataset can resolve. The interaction in the fineness $\times$ diesel plane is not an exception, since it displaces both grades together.

The pulp measurements indicate where selectivity is instead determined. The concentrator faces a colder pulp and a different clay loading in winter, and the two act in opposing directions: heating lowers the viscosity penalty but also lowers the contact angle of a clean concentrate and raises that of one carrying chlorite. Kaolinite, both the most viscous clay and the one whose viscosity does not respond to temperature, cannot be assisted by heating. Converting this mechanism into a control strategy requires flotation tests at controlled temperature with the clay mineralogy characterised.

The data come from a single deposit, so the numerical results are local, and reagent–reagent interactions are unidentified.